Healthcare Data Interoperability Policy Analysis

Request for Information

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HEALTHCARE DATA INTEROPERABILITY POLICY ANALYSIS

At Cognitive, we have witnessed the profound disconnect between the promise and reality of healthcare data interoperability, a gap that reflects not just technical challenges but also fundamental misalignments in how we approach healthcare transformation. Despite more than a decade of federal investment and regulatory mandates, the vision of seamless, patient-centered health information exchange remains frustratingly elusive, trapped between ambitious policy frameworks and the messy realities of healthcare delivery, where competing priorities, resource constraints, and entrenched workflows create implementation barriers that no amount of regulatory language can overcome.

Introduction

The pursuit of healthcare data interoperability has become both our industry's greatest opportunity and its most persistent source of frustration, representing a complex challenge that extends far beyond technical specifications and touches every aspect of how we deliver, coordinate, and experience healthcare in America. Cognitive has learned that interoperability isn't just a technical issue, but a fundamentally a human problem disguised as a technical challenge. The most sophisticated FHIR APIs and blockchain solutions mean nothing if they do not translate into better experiences for patients navigating complex health conditions or clinicians trying to deliver coordinated care.

Current Policy Goals and Objectives. Federal agencies have articulated remarkably comprehensive visions for healthcare data interoperability that address every stakeholder's needs on paper, yet these policy objectives often exist in isolation from the messy realities of healthcare delivery, where the disconnect between policy aspiration and operational reality has created a landscape where technical compliance frequently substitutes for meaningful progress. The most troubling aspect we see is the tendency to treat interoperability as a destination rather than a journey, with regulations specifying technical requirements and compliance timelines while providing limited guidance on navigating the complex organizational, financial, and cultural changes necessary to achieve meaningful interoperability.

Centers for Medicare & Medicaid Services (CMS) Strategic Framework. CMS has established a comprehensive strategic framework that recognizes interoperability as fundamental to healthcare transformation, promoting secure health information exchange to support better decision-making and system efficiency with their vision of "a secure, connected health care system that empowers patients and their providers" reflecting sophisticated understanding of interoperability as an enabler of broader healthcare goals rather than an end in itself. However, the gap between CMS's strategic vision and practical implementation reveals the limitations of top-down policy approaches, where the agency's emphasis on FHIR APIs as the technical foundation, while technically sound, can become an obstacle when organizations focus on compliance rather than improving healthcare delivery and patient outcomes.

Office of the National Coordinator (ONC) Regulatory Initiatives. The Office of the National Coordinator has pursued interoperability primarily through regulatory mechanisms stemming from the 21st Century Cures Act, focusing on empowering patients with medical data access and addressing information blocking through mandatory compliance requirements that represent a

significant shift from voluntary adoption to regulatory enforcement. What concerns us at Cognitive about ONC's regulatory approach is its emphasis on preventing bad behavior rather than incentivizing good outcomes. While information blocking regulations are necessary, they don't adequately address the more common challenges of technical limitations, resource constraints, and workflow incompatibilities that effectively prevent meaningful data sharing.

Department of Health and Human Services (HHS) Comprehensive Data Strategy. HHS has developed a comprehensive data strategy for 2023-2028 that extends beyond healthcare-specific interoperability to encompass broader HHS data integration, identifying five critical priority areas including cultivating data talent, fostering data sharing, and enabling whole-person care delivery, with explicit connections to Cancer Moonshot initiative demonstrating how data interoperability serves broader public health objectives. The challenge we see at Cognitive is that while this comprehensive approach is conceptually sound, it requires levels of coordination and data integration that current healthcare IT infrastructure and organizational capabilities cannot support, suggesting that the strategy's identification of preparedness and incident response as anchor use cases reflects recognition that crisis situations often reveal both the potential and the limitations of our current interoperability capabilities.

Policy Gaps and Shortcomings. The most significant policy gaps we observe at Cognitive stem from the disconnect between regulatory aspirations and implementation realities across diverse healthcare settings, where federal policies work reasonably well for large, well-resourced organizations but create compliance burdens without corresponding benefits for smaller practices, specialty providers, and safety-net organizations. Perhaps more concerning is the framework's limited attention to user experience and patient engagement, where regulations specify technical requirements but provide minimal guidance on making these capabilities meaningful and accessible to patients, as evidenced by consumer research revealing that over half of healthcare consumers find accessing medical records "stressful and time consuming”.

Implementation Disparities Across Healthcare Settings. The reality of interoperability implementation reveals stark disparities that undermine the vision of universal health information exchange, with only 67% of psychiatric hospitals achieving certified electronic health record (EHR) adoption compared to 86% of general acute care hospitals, while rural healthcare settings and long-term care facilities face particularly acute challenges that current policy frameworks inadequately address. These implementation disparities often reflect deeper organizational and financial challenges that technical solutions alone cannot address, creating a fragmented healthcare landscape where patients moving between different care settings encounter information silos that compromise care coordination and safety.

Patient Access Barriers and User Experience Challenges. Consumer research reveals a troubling disconnect between regulatory requirements for patient data access and actual patient experiences. As stated above, over half of healthcare consumers reporting that accessing medical records is "stressful and time consuming," while more than 40% express hesitation to switch providers for fear of losing access to records and data. The fundamental design limitations of patient portal systems, the primary mechanism for consumer health data access, reveal how technical compliance can do little for user experience, with half of study participants reporting that portals "aren't user friendly" and 63% experiencing problems with getting historical records.

Fragmentation in Health Information Exchange Participation. The voluntary nature of health information exchange participation has created a patchwork of connectivity that fundamentally limits the effectiveness of interoperability policies, with different HIE models operating with varying technical standards, governance structures, and participation requirements that create a fragmented landscape where even providers with certified EHR systems may not be able to exchange data with all relevant care partners. The technical heterogeneity created by EMR vendor implementations compounds these connectivity challenges, with hundreds of EMR systems each having their own technical specifications, clinical terminologies, and unique customizations that prevent true interoperability and data sharing across systems.

Technology Infrastructure Requirements and Standards. The technology infrastructure requirements embedded in current interoperability policies represent both significant achievements and persistent limitations in our approach to healthcare data exchange, with regulations such as the HTI-1 final rule establishing unprecedented transparency requirements for artificial intelligence while revealing the reactive nature of healthcare IT policy that struggles to keep pace with technological innovation. At Cognitive, we've learned that successful technology implementation requires not just technical specifications but also adequate time, training, and support to help organizations navigate complex transitions without compromising care delivery or operational stability.

HTI-1 Final Rule: Algorithmic Transparency and Certification Updates. The HTI-1 final rule represents a significant advancement in addressing the emerging intersection of artificial intelligence and healthcare interoperability, establishing unprecedented transparency requirements for AI and predictive algorithms embedded in certified health information technology that recognize the need for accountability mechanisms enabling clinical judgment and patient advocacy. However, the challenge we see at Cognitive is that transparency requirements alone may not be sufficient to address the complex ethical and practical challenges raised by healthcare AI, as healthcare organizations need guidance, training, and support to effectively evaluate and implement these technologies in ways that improve rather than complicate clinical workflows.

HTI-2 Final Rule: TEFCA Implementation and Regulatory Framework. The HTI-2 final rule establishes the regulatory foundation for the Trusted Exchange Framework and Common Agreement (TEFCA), creating new federal regulations that provide greater transparency and implement provisions to support reliability, privacy, security, and trust within TEFCA, representing a significant step toward standardized, nationwide health information exchange. The rule's creation of the TEFCA Manner Exception represents a significant policy development that could either facilitate or constrain data sharing depending on implementation, as it allows healthcare providers and developers to limit fulfillment of health information requests to only through TEFCA, which aims to create standardized pathways but raises concerns about potentially limiting patient choice.

FHIR API Implementation and Standards Evolution. The widespread adoption of FHIR APIs represents the most significant technical advancement in healthcare interoperability policy implementation, with CMS regulations requiring payers to implement Patient Access APIs, Provider Directory APIs, and Payer-to-Payer APIs using FHIR standards, which enable a fundamental shift from proprietary data exchange methods to standardized, patient-controlled access mechanisms. However, FHIR implementation has revealed significant challenges related

to data quality, completeness, and standardization that technical specifications alone cannot address, as the underlying clinical data often lacks the consistency and completeness necessary for effective care coordination, with variations in clinical terminology, data entry practices, and system configurations continuing to create barriers to meaningful information exchange.

Impact on Value-Based Care and Patient-Focused Access. The transition to value-based care models requires comprehensive patient data integration that extends beyond traditional clinical information to encompass social determinants of health, care coordination activities, and outcome measurements, with interoperability serving as "the backbone of successful value-based care initiatives," yet current implementations often fall short of providing the comprehensive data integration necessary for effective value-based care management. Cognitive has observed how the fragmented nature of current health information systems creates particular challenges for value-based care organizations serving complex patient populations, where the requirement to integrate data from multiple sources often requires manual data reconciliation processes that are both time-consuming and error-prone.

Value-Based Care Integration Challenges. Value-based care models require a level of data integration and care coordination that current interoperability implementations struggle to support effectively, extending beyond technical connectivity to encompass data quality, clinical workflow integration, and outcome measurement capabilities that necessitate sophisticated health IT infrastructure and organizational capabilities. The emphasis on API-based data access, while providing technical connectivity, doesn't automatically address the care coordination challenges that are central to value-based care success, creating a gap between technical data availability and practical care delivery integration that continues to limit the effectiveness of these models.

Patient Access Implementation Gaps. Despite regulatory requirements for patient data access, implementation has revealed significant limitations in both technical functionality and user experience that undermine the vision of patient-centered care, with the MyHealthEData initiative's goal of giving "every American control of their medical data" remaining largely unrealized in practical healthcare delivery. The challenge is compounded by patient access to health information, which often does not extend to the complete longitudinal health record that would be most valuable for patient decision-making and care coordination. Many portal systems provide access only to information generated within specific healthcare organizations, rather than comprehensive health histories.

Persistent EMR and Data Sharing Challenges. Electronic Medical Record systems continue to present fundamental architectural barriers to effective data sharing despite compliance with interoperability standards, with the proliferation of EMR customizations and proprietary extensions creating a landscape where hundreds of systems each have their own technical specifications and unique customizations that prevent true interoperability. The focus on EMR adoption rather than interoperability during the initial HITECH Act implementation established a foundation of systems not designed with comprehensive data sharing as a primary objective, while data quality and standardization issues represent persistent challenges that technical standards alone cannot address.

Technical Architecture Limitations. The technical architecture limitations of current EMR systems reflect fundamental design decisions made during an era when interoperability was not a

primary consideration, with EMR vendors implementing systems optimized for internal organizational workflows rather than external data sharing, creating architectural barriers that persist despite subsequent regulatory requirements. Cognitive has seen firsthand that these limitations create ongoing challenges for healthcare organizations trying to implement meaningful interoperability, as the volume of data produced by healthcare IT systems often leads to information overload rather than actionable clinical intelligence.

Policy Framework Inadequacies. Current regulatory frameworks, although comprehensive in scope, contain significant gaps that limit their effectiveness in addressing real-world interoperability challenges. Information blocking regulations primarily focus on intentional data withholding but do not adequately address technical limitations, resource constraints, or workflow incompatibilities that may effectively prevent data sharing. The regulatory emphasis on standardized APIs and technical specifications has not been accompanied by sufficient attention to implementation support, user education, and workflow integration, creating a gap between technical compliance and operational effectiveness that continues to limit the practical impact of interoperability policies.

Patient Experience Frustrations. Patient experiences with healthcare data access reveal persistent frustrations that indicate fundamental limitations in current interoperability implementations, with consumers reporting that accessing medical records remains "stressful and time consuming" while many experience repeated testing or cancelled appointments due to lack of access to test results. Patients often find themselves serving as the primary mechanism for information transfer between providers, undermining the vision of seamless electronic health information (EHI) exchange, while patient portal systems fail to present information in ways that support decision-making and care management, with medical terminology and fragmented information presentation creating barriers to meaningful engagement.

Current Federal and Industry Position on Patient-Controlled Health Records vs. Distributed Model. The federal government and healthcare industry maintain complex positions regarding patient control versus distributed health record models, with policy rhetoric emphasizing patient empowerment while practical implementation reinforces organizational custody of patient health information, as exemplified by the MyHealthEData initiative promising patient control while maintaining structures that keep authoritative records within provider systems. Current federal regulations focus primarily on providing patient access to copies of health information rather than enabling patients to maintain authoritative health records, treating patients as consumers rather than active participants in health record creation and maintenance, while industry resistance stems from liability concerns, regulatory compliance requirements, and business model considerations. Industry resistance to patient-controlled health records stems from multiple factors including liability concerns, regulatory compliance requirements, and business model considerations that create financial incentives for organizations to maintain control over patient data as a competitive advantage.

Rationale for Provider-Controlled Records. Healthcare organizations justify maintaining control over authoritative health records based on arguments related to clinical responsibility, data integrity, and regulatory compliance that reflect legitimate concerns about patient safety and care quality, with providers arguing they bear ultimate responsibility for clinical decisions requiring control over supporting health information. Regulatory compliance requirements create practical

barriers to patient-controlled health records, as healthcare organizations must meet numerous federal and state requirements related to record retention, privacy protection, and audit trails that are more easily managed within organizational systems providing centralized control over data governance.

Emerging Technologies: Blockchain and Self-Sovereign Identity. Blockchain technology presents promising solutions to fundamental challenges limiting patient-controlled health records, particularly related to data integrity, security, and interoperability, with blockchain-based systems enabling patients to maintain authoritative health records while providing cryptographic verification and comprehensive audit trails. Self-sovereign identity technologies offer complementary capabilities that could address authentication, authorization, and consent management challenges, enabling individuals to store and manage health data in secure digital wallets with granular control over access permissions, though these technologies also present significant implementation challenges, including energy consumption concerns and the need for new technical infrastructure.

Extended Episodes of Care Models. Extended episodes of care models represent a fundamental shift from traditional fee-for-service approaches to comprehensive value assessment encompassing all healthcare actors and factors contributing to patient outcomes, requiring sophisticated data integration capabilities that extend far beyond current interoperability implementations to capture social determinants of health, care coordination activities, and long-term outcome measurements. Current interoperability standards provide some foundation for these models but fall short of the comprehensive data integration required, while policy frameworks must address attribution challenges, outcome measurement standardization, and risk adjustment methodologies that account for the complexity of factors influencing health outcomes.

Technical and Policy Requirements for Comprehensive Care Valuation. Extended episodes of care models require sophisticated data integration capabilities that extend far beyond current interoperability implementations to capture social determinants of health, care coordination activities, patient behaviors, and long-term outcome measurements, with technical infrastructure needs including real-time data integration from multiple sources, predictive analytics capabilities, and outcome tracking systems operating across extended time periods. Policy frameworks for extended episodes must address attribution challenges, outcome measurement standardization, and risk adjustment methodologies accounting for the complexity of factors influencing health outcomes, requiring new regulatory frameworks that can accommodate multi-sector care coordination complexity while maintaining accountability for outcomes and costs.

Innovation Requirements for Comprehensive Care Coordination. The implementation of extended episodes of care models requires technological innovations that extend beyond current health IT capabilities to encompass care coordination, patient engagement, and outcome prediction tools. Artificial intelligence and machine learning technologies are essential for processing vast amounts of data and identifying care optimization patterns. Blockchain and distributed ledger technologies could play a crucial role by providing secure, auditable mechanisms for tracking care contributions, outcome attribution, and payment distribution across multiple organizations and time periods. Meanwhile, patient engagement technologies and self-sovereign identity solutions enable patients to maintain control over their health information while participating in complex care coordination activities.

Conclusion. The analysis of current healthcare data interoperability policies reveals a complex landscape of ambitious objectives, significant achievements, and persistent challenges that continue to limit the realization of patient-centered, interoperable healthcare delivery, with fundamental gaps remaining between technical compliance and meaningful healthcare transformation despite comprehensive federal policy frameworks and substantial regulatory requirements. We believe that future progress will require more fundamental changes to healthcare delivery models, including greater adoption of patient-controlled health records enabled by blockchain and self-sovereign identity technologies, comprehensive care coordination systems integrating social determinants of health, and extended episodes of care models aligning incentives with comprehensive patient outcomes—innovations requiring not only technical advancement but also new regulatory frameworks, incentive structures, payment models, and organizational approaches that truly center patient needs and preferences in healthcare delivery and data management.

COGNITIVE RESPONSE TO SELECT QUESTIONS

PC-1. What health management or care navigation apps would help you understand and manage your (or your loved ones) health needs, as well as the actions you should take?
a. What are the top things you would like to be able to do for your or your loved ones' health that can be enabled by digital health products?
b. If you had a personal assistant to support your health needs, what are the top things you would ask them to help with? In your response, please consider tasks that could be supported or facilitated by software solutions in the future.

Patients and caregivers often struggle with fragmented, inaccessible health data spread across multiple portals and systems. A truly effective, national, digital health services platform would unify all medical records, from lab results and imaging studies to specialist notes and medication histories, into a single, intuitive interface that is directly accessible by the patient at no cost. This platform should be capable of both on-demand and automatic syncing with every healthcare provider's system, eliminating the need for patients to manually request or upload records. For example, after a hospital discharge, the app would proactively aggregate rehab plans, new prescriptions, and follow-up instructions into a timeline, flagging critical steps such as "Schedule cardiac rehab within 14 days" or "Check potassium levels weekly." Data received from non-self-systems should be labeled as such and meta data for each received data element should have its complete provenance documented. Once current EMR vendors have the technical ability to distinguish "self-generated" data from third-party data, patients must be allowed to enter their own data or data that is currently not well documented in any existing EMR, to include, OTC medications, vitamins, health goals, advanced directives, etc.

A top patient request is context-aware medication management and automated reconciliation. Current apps offer basic reminders, but users want systems that explain "why" a drug matters ("This beta-blocker lowers your heart attack risk by 30%"), warn about conflicts ("Avoid grapefruit with this statin"), and auto-refill prescriptions by analyzing usage patterns. For caregivers managing complex regimens, dose-by-dose tracking with AI-driven alerts (e.g., "Missed insulin dose detected via connected pen") could prevent dangerous oversights.

Patients also desire intelligent care coordination tools. Imagine a "health assistant" that books appointments based on provider availability, insurance coverage, and care gaps (e.g., "You're due for a mammogram—here are 3 in-network clinics with Saturday slots"). For loved ones, shared task boards could assign responsibilities such as "Uncle Jim: Drive to chemo every Tuesday" or "Sister Maya: Monitor Dad's glucose readings." The app would facilitate secure communication with care teams, translating medical jargon into plain language (e.g., "Your LDL is 120—let's discuss lowering it below 100").

Proactive health guardianship is another unmet need. Advanced apps could analyze wearable data, EHR trends, and SDOH factors (e.g., local air quality) to predict risks. A parent might receive: "Asthma alert: Pollen levels spiking tomorrow. Pack the inhaler for school and avoid outdoor recess." For seniors, fall detection sensors could auto-alert family and 911 while displaying critical data to EMTs (e.g., "Allergies: Penicillin; Current medications: Warfarin").

If given a "health assistant," patients would prioritize crisis-ready functionality. During emergencies, the app should instantly share living wills, medication lists, and provider contacts

with ER teams. For daily needs, it could automate insurance paperwork, dispute billing errors using AI, and even negotiate payment plans. Crucially, all features must prioritize universal accessibility, including voice commands for those with limited dexterity, offline mode for rural users, and multilingual support to bridge language gaps.

These tools address core frustrations: disjointed data, passive systems that wait for user input, and apps that add complexity rather than reducing it. By centralizing, contextualizing, and automating care navigation, digital health can finally meet patients where they are: overwhelmed but eager to take control.

PC-4. What features are missing from apps you use or that you are aware of today?

a. What apps should exist but do not yet? Why do you believe they do not exist yet?

b. What set of workflows do you believe CMS is uniquely positioned to offer?

A critical gap in current digital health tools is the absence of truly unified patient portals that transcend institutional boundaries. While apps such as MyChart allow access to records within a single health system, they fail to aggregate data from Medicare, Medicaid, private insurers, and specialty providers into a cohesive timeline. Patients managing chronic conditions often spend hours reconciling conflicting medication lists from different EHRs, while caregivers struggle to track appointments across unconnected provider portals. This fragmentation persists because EHR vendors prioritize loyalty to health systems over patient needs, maintaining proprietary data formats that resist integration. CMS is uniquely positioned to mandate a Federal Health Data Aggregation Standard, requiring all federally funded providers to contribute real-time updates to a government-hosted patient portal using FHIR R4.0.1. By leveraging its authority under the 21st Century Cures Act, CMS could compel vendors such as Epic and Cerner to auto-populate this portal with structured clinical data, claims histories, and SDOH indicators, without charging patients for access. Recognizing that achieving such a goal would be a multi-year process, CMS should mandate that that all data be semantically constrained using publicly accessible vocabularies. Vendors currently unable to deliver completely structured data, such as notes, should be required to provide context appropriate wrappers around the unstructured data to facilitate post-receipt NLP and processing.

Another glaring omission is the lack of AI-powered care navigation tools that contextualize health data for lay users. Existing apps might show lab results but rarely explain their implications (e.g., “Your A1C of 8.2 indicates poorly controlled diabetes—here’s how to adjust your diet”) or automate next steps (e.g., scheduling retinal exams when glucose levels rise). These tools don’t exist because developers face insurmountable barriers accessing comprehensive, real-time patient data across disparate EHRs. CMS could solve this by expanding its Blue Button 2.0 API into a National Health Data Utility, ingesting Medicare/Medicaid claims, provider-generated EHR data, and wearable metrics into a secure repository that, again, patients would have no-cost access to. Third-party apps that meet strict federal certification criteria (e.g., non-commercial use, accessibility compliance) could then query this utility through standardized APIs, enabling features such as predictive hypoglycemia alerts for diabetics or fall-risk assessments for seniors, all while maintaining patient privacy through NIST IAL2 credentialing.

CMS should also pioneer crisis-ready health profiles that automatically share critical data during emergencies. Current apps rely on patients to manually input allergies, medications, and advance

directives, a process prone to errors and omissions. By contrast, a CMS-mandated emergency profile could pull verified data from the FHIR-based national portal and transmit it securely to EMS systems via QR codes or NFC chips embedded in Medicare cards. This would address the 34% of medical errors caused by incomplete information during emergencies. Resistance from EHR vendors, who profit from siloed data and the resale of de-identified patient data, makes this untenable without federal intervention. CMS could tie Medicare reimbursement to real-time emergency data sharing, penalizing providers who delay updates.

For underserved populations, offline-first health tools are urgently needed. Rural patients without reliable broadband often can't use app features requiring constant connectivity, exacerbating disparities in care. CMS could deploy federally funded health kiosks in post offices and libraries, offering offline access to medication histories, telehealth visits, and appointment scheduling via USB-C health keys. These kiosks would regularly sync data when internet access is available, ensuring continuity. Private vendors avoid such solutions due to low profit margins, but CMS could fund them through Medicaid IT modernization grants.

Finally, CMS must address systemic inequities in app design. Many tools lack voice navigation, low-literacy interfaces, or ASL support, all of which are barriers that contradict the agency's mission. By establishing mandatory accessibility certifications for any app accessing federal health data, CMS could force developers to prioritize universal design. This aligns with the HHS Section 508 mandates but extends them to private-sector tools used by Medicare beneficiaries. Without such requirements, market-driven solutions will continue excluding those most in need of care coordination.

These solutions require CMS to act as architect, regulator, and steward—roles that only a federal agency can fulfill at scale. By wielding its purchasing power (e.g., Medicare's coverage of 60 million lives) and regulatory authority, CMS can dismantle vendor silos, enforce equity, and deliver the seamless, patient-centered health ecosystem envisioned in the 21st Century Cures Act.

PC-5. What can CMS and its partners do to encourage patient and caregiver interest in these digital health products?

a. What role, if any, should CMS have in reviewing or approving digital health products on the basis of their efficacy, quality or impact or both on health outcomes (not approving in the sense of a coverage determination)? What criteria should be used if there is a review process? What technology solutions, policy changes, or program design changes can increase patient and caregiver adoption of digital health products (for example, enhancements to data access, reimbursement adjustments, or new beneficiary communications)?

b. What changes would enable timely access to high quality CMS and provider generated data on patients? CMS role in reviewing products:

To catalyze patient and caregiver engagement with digital health tools, CMS must use its regulatory authority to dismantle systemic barriers while mandating patient-centric innovation from EHR vendors. A cornerstone of this approach is universal, free access to interoperable health apps certified through a rigorous federal review process. CMS should require all federally funded providers to offer patients a government-developed patient portal that aggregates records from every care site (such as hospitals, specialists, and/or labs) into a single interface. This portal would

correct current fragmented systems by enforcing FHIR R4.0.1 standards across all vendors, ensuring seamless data integration regardless of the originating EHR. For rural patients lacking broadband, CMS could partner with the FCC to deploy offline-enabled kiosks at post offices and community centers, granting equitable access to health data and telemedicine services.

This government-developed patient portal should not be implemented simply as a read-only portal, but rather as a digital health platform upon which bidirectional services and analytic capabilities can be deployed. A universal, national data interoperability platform is but the first step. This platform must eventually develop into a national knowledge management platform upon which advanced analytics, AI and automated quality assurance systems can be built. Current health IT infrastructure should be viewed as a basic utility and priced accordingly. Only when we can focus on the knowledge that drives the healthcare domain, will truly exponential advances in outcomes be achieved.

CMS' role in product review should focus on outcome-based certification rather than gatekeeping coverage. Drawing from the Digital Quality Measurement Strategic Roadmap, the agency could establish a National Health App Validation Program assessing tools on:

- Clinical Impact: Demonstrated reduction in hospitalizations or improved chronic disease management via peer-reviewed studies.
- Interoperability Dept: Ability to ingest/export 100% of USCDI v3 elements, including SDOH data, via standardized APIs.
- Accessibility: Compliance with Section 508 standards, including voice navigation and low-literacy interfaces.
- Security: NIST IAL2/AAL2 credential integration and blockchain-backed audit trails to prevent breaches.

To drive adoption, CMS should recalibrate value-based payments to reward providers who achieve >75% patient activation in certified apps. Simultaneously, the agency could launch Medicare Digital Health Credits, such as $500 annual allowances for beneficiaries to purchase CMS-approved devices (e.g., glucose monitors, tablets) or offset internet costs. This mirrors successful broadband subsidies but targets health-specific needs.

For data liquidity, CMS must expand Blue Button 2.0 into a real-time health data utility, mandating EHR vendors to push updates to patient apps within 24 hours of provider documentation. The HTI-2 Final Rule's patient access provisions provide a regulatory hook for this requirement. Parallel efforts should unify Medicare Advantage (Part C) claims with traditional Medicare data, closing gaps that currently force patients to manually reconcile disparate records.

To ensure data quality, CMS could deploy AI validation bots that cross-check EHR entries against claims and patient-reported outcomes. Providers with >95% accuracy scores would receive streamlined quality reporting requirements, while those below thresholds face mandatory EHR workflow audits. This creates a virtuous cycle where data integrity becomes a shared priority.

Finally, CMS must lead a national digital health literacy campaign, embedding training modules into Medicare Annual Wellness Visits and partnering with organizations such as AARP to host community workshops. By positioning itself as both regulator and educator, CMS can transform

passive beneficiaries into empowered health consumers while compelling vendors to prioritize public good over proprietary interests.

PC-7. If CMS were to collect real-world data on digital health products' impact on health outcomes and related costs once they are released into the market, what would be the best means of doing so?

To systematically capture real-world data (RWD) on digital health products' outcomes and costs, CMS must leverage its regulatory authority to mandate a National Digital Health Evidence Platform, integrated with the Trusted Exchange Framework and Common Agreement (TEFCA) and Qualified Health Information Networks (QHINs). This platform would serve as a centralized hub, requiring all certified digital health tools regardless of vendor to automatically report de-identified patient outcomes, utilization patterns, and cost data via FHIR R4.0.1 APIs. By embedding this requirement into TEFCA's governance framework, CMS can ensure that every interaction with a digital health product (e.g., medication adherence alerts, telehealth visits, remote monitoring alerts) generates structured data streams accessible to researchers, policymakers, and even patient's themselves where appropriate.

The platform would aggregate three core data types:

- Clinical Outcomes: Pulled directly from EHRs via QHINs, including changes in biometrics (e.g., HbA1c reductions), hospital readmissions, and adverse events.
- Patient-Reported Data: Captured through CMS-certified apps using standardized PROMIS® measures, with voice-to-text functionality for low-literacy users.
- Cost/Utilization Metrics: Tied to Medicare claims via an expanded Blue Button 3.0 API, tracking downstream savings from reduced ER visits or avoided complications.

To ensure rigor, CMS could adapt the HARPER+ protocol, a structured template for RWD studies, into an automated validation engine. This system would flag anomalies (e.g., mismatches between app-reported adherence and prescription fills) and require vendors to submit root-cause analyses. Crucially, the platform would integrate with CMS's proposed national provider directory, enabling attribution of outcomes to specific providers and technologies while maintaining patient privacy through NIST IAL2/AAL2 digital credentials.

Challenges such as EHR fragmentation and proprietary data silos would be addressed by mandating participation for all TEFCA-certified entities. For example, Epic and Cerner would be required to expose FHIR endpoints for real-time data extraction, with financial penalties for non-compliance. Simultaneously, CMS could deploy AI-powered audit bots to cross-validate submitted data ensuring fidelity.

To incentivize adoption, CMS could:

- Link Medicare Advantage Star Ratings to digital health tool utilization and outcome transparency.
- Offer shared savings bonuses to Accountable Care Organizations (ACOs) that achieve >80% patient enrollment in certified apps.
- Waive prior authorization requirements for providers using CMS-validated technologies with proven cost reductions.

This approach transforms TEFCA from a passive data conduit into an active evidence-generation engine, aligning with the Digital Quality Measurement Strategic Roadmap's vision of FHIR-based measure calculation tools. By centralizing infrastructure and enforcing interoperability, CMS avoids replicating the failures of vendor-driven platforms that prioritize profit over patient impact, ensuring that real-world evidence becomes a public good, not a proprietary asset.

PC-8. In your experience, what health data is readily available and valuable to patients or their caregivers or both?

a. What data is valuable, but hard for patients and caregivers, or app developers and other technical vendors, to access for appropriate and valuable use (for example, claims data, clinical data, encounter notes, operative reports, appointment schedules, prices)?

b. What are specific sources, other than claims and clinical data, that would be of highest value, and why?

c. What specific opportunities and challenges exist to improve accessibility, interoperability and integration of clinical data from different sources to enable more meaningful clinical research and generation of actionable evidence?

Patients and caregivers currently grapple with fragmented access to health data that fails to meet their needs for comprehensive care coordination. While basic claims data (e.g., Medicare Parts A/B/D via Blue Button 2.0) and limited clinical data (e.g., lab results, medication lists) are technically accessible, their utility is undermined by systemic silos. For example, a patient discharged after cardiac surgery might access their hospital's portal for operative notes but find their post-acute rehab plan locked in a separate long-term care EHR. Caregivers managing elderly parents face even greater hurdles, often acting as human intermediaries to manually compile records from specialists, pharmacies, and home health agencies.

High value but inaccessible data is rampant throughout the system, and includes, but is not limited to, clinical encounter notes, operative reports, pathology staging reports, orders and real-time appointment schedules and statuses. These are often trapped in proprietary EHR systems that resist interoperability, despite their critical role in care continuity. A caregiver trying to coordinate a dementia patient's neurology follow-up might spend hours calling offices to obtain appointment slots, while a diabetic patient' continuous glucose monitor data remains isolated from their PCP's EHR. Price transparency data, essential for avoiding surprise bills, is similarly fragmented, buried in disparate payer portals or PDF fee schedules.

Beyond claims and clinical data, social determinants of health (SDOH) and wearable device metrics represent untapped goldmines. SDOH factors such as housing stability or food insecurity, documented in community health databases, rarely flow into clinical workflows. Wearables capturing real-time heart rate variability or sleep patterns remain disconnected from care teams, forcing patients to verbally recount self-reported data, which providers often distrust. These gaps persist because current policies prioritize clinical over holistic data, and vendors lack incentives to integrate non-reimbursable metrics.

Significant opportunities for improvement hinge on CMS leveraging its regulatory authority to redefine data accessibility. By mandating FHIR-based SDOH fields in the USCDI v4 and requiring EHR vendors to ingest community-based data via TEFCA QHINs, CMS could bridge clinical and

social care. For example, a Medicaid patient's housing instability documented by a local food pantry could automatically trigger alerts in their primary care EHR, enabling targeted interventions. Similarly, real-world data from wearables and remote monitoring tools could feed into a national evidence-generation platform, akin to Medidata's Integrated Evidence Solutions, to accelerate research on chronic disease management.

Challenges are rooted in legacy systems and misaligned incentives. Many hospitals still rely on HL7v2 interfaces that cannot support modern FHIR APIs, while EHR vendors profit from data lock-in via proprietary extensions. CMS must counter this by:

- Conditioning Medicare reimbursement on demonstrated FHIR R4.0.1 compliance, including bidirectional SDOH data exchange.
- Funding rural HIEs to aggregate community health worker reports, school-based clinic records, and aging-in-place sensor data into TEFCA-aligned formats.
- Deploying AI validation bots to audit EHR-to-QHIN data flows, ensuring critical elements such as advance directives or allergy lists are not omitted during exchange.

A national health data utility built on TEFCA's backbone could democratize access while preserving privacy. Imagine a caregiver logging into a CMS-hosted portal to see their child's asthma action plan from the school nurse, recent corticosteroid refills from the pharmacy, and pollen forecasts from EPA APIs all contextualized by an AI assistant flagging high-risk days. Such integration is feasible today but requires CMS to compel vendors to abandon walled gardens in favor of collective good.

In conclusion, the path forward demands CMS act as architect and enforcer, using TEFCA/QHIN mandates to unify clinical, SDOH, and consumer-generated data into one infrastructure. Only through centralized governance can we transform today's disparate data scraps into the seamless, patient-controlled health narrative that empowers individuals and advances population health.

PC-10. How is the Trusted Exchange Framework and Common Agreement™ (TEFCA™) currently helping to advance patient access to health information in the real world?
a. Please provide specific examples.
b. What changes would you suggest?
c. What use cases could have a significant impact if implemented through TEFCA?
d. What standards are you aware of that are currently working well to advance access and existing exchange purposes?
e. What standards are you aware of that are not currently in wide use, but could improve data access and integration?
f. Are there redundant standards, protocols, or channels that should be consolidated?
g. Are there adequate alternatives outside of TEFCA for achieving widespread patient access to their health information?

The Trusted Exchange Framework and Common Agreement (TEFCA) represents the federal government's most ambitious effort to dismantle healthcare data silos and empower patients. By establishing a "network of networks" through Qualified Health Information Networks (QHINs), TEFCA has begun connecting disparate health systems, payers, and public health agencies under a unified interoperability framework. Early successes include demonstrations such as the 2025 HIMSS showcase, where patients used TEFCA-aligned apps to retrieve longitudinal health records

across states, reducing care delays for those with complex conditions such as rare diseases or cancer. Public health agencies also benefit: the CDC Foundation's "eCR over TEFCA" initiative enabled 58 jurisdictions to receive 1 million+ electronic case reports for disease surveillance, illustrating TEFCA's potential to modernize public health infrastructure. However, these wins remain localized, hindered by voluntary participation and inconsistent implementation.

To fulfill its promise, CMS must transition TEFCA from an opt-in framework to a mandatory national utility, setting the stage for a national infrastructure to support true knowledge management services. This requires:

- Universal Participation Mandates: Conditioning Medicare/Medicaid reimbursement on TEFCA compliance for all providers and payers, closing gaps caused by voluntary adoption.
- Patient-Centric Governance: Implementing principles from the JMIR analysis (2022) to grant patients granular control, including audit logs of data access and FHIR-based consent management tools.
- Federal Identity Resolution: Addressing patient matching failures (a persistent issue noted in Medplum's TEFCA guide) by deploying a national MPI backed by NIST IAL2 digital credentials, eliminating reliance on error-prone demographic matching.

Use cases which could have a significant impact if implemented through TEFCA include:

- Crisis Readiness: Automating emergency access to advance directives, medication lists, and specialist contacts during 911 calls, as demonstrated by Health Gorilla's QHIN platform.
- Chronic Care Coordination: Enabling apps to aggregate data from primary care, pharmacies, and wearable devices into AI-driven care plans, alerting providers when patients miss dialysis or HbA1c levels spike.
- Price Transparency: Integrating CMS-mandated payer pricing APIs with TEFCA to let patients compare procedure costs across in-network providers pre-service.

FHIR R4.0.1 and USCDI v3 form TEFCA's technical backbone, but legacy systems still rely on outdated protocols such as HL7v2 and CCDA, which lack granular patient consent controls. While Carequality and CommonWell have aligned with TEFCA, proprietary variations persist. Epic's FHIR implementation, for example, often requires custom extensions. To resolve this, CMS should:

- Retire Redundant Standards: Sunset HL7v2 and CCDA in favor of FHIR Document Reference for summaries.
- Enforce USCDI+ Adoption: Mandate inclusion of SDOH data (e.g., housing status, food insecurity), consent directives, patient goals, and patient-reported outcomes in all exchanges.
- Expand SMART on FHIR: Require EHR vendors to support this standard for third-party app integration, reducing reliance on proprietary app galleries.

While vendor-specific networks such as Epic's Care Everywhere process billions of annual transactions, they prioritize institutional loyalty over patient.

PR-1. What can CMS and its partners do to encourage providers, including those in rural areas, to leverage approved (see description in PC-5) digital health products for their patients?
a. What are the current obstacles?
b. What information should providers share with patients when using digital products in the provision of their care?
c. What responsibilities do providers have when recommending use of a digital product by a patient?

The average rural provider in the US, is a less-than-five-person organization, office, or clinic that has a fax machine, an internet connection, a billing system and perhaps a simple practice management system. There are no current incentives in place for this type of organization to adopt an EHR system, integrate with an HIE or adopt a FHIR-enabled interoperability platform for exchanging information. The lowest common denominator technology available to this type of office is the fax machine. CMS should expand financial incentives for rural providers to adopt interoperable digital tools, such as subsidizing EHR upgrades and offering bonus payments under value-based care models. Technical assistance programs, including FHIR API implementation support and workflow redesign consulting, could address integration challenges. CMS should also provide incentives for technology providers to target this marketspace.

- Current obstacles: Rural providers face high upfront costs for certified EHR systems and insufficient broadband infrastructure, with 25% of rural hospitals lacking reliable internet access. Interoperability gaps between legacy systems and modern APIs create double documentation burdens, as seen in practices manually entering data across non-connected platforms.
- Information to share with patients: Providers must explain data privacy protections, such as HIPAA app compliance, and provide written instructions for accessing/using recommended tools. Clear disclosures about data usage rights, such as third-party app data monetization policies, are essential.
- Provider responsibilities: Clinicians should vet apps for clinical validity (e.g., FDA clearance for diagnostic tools) and ensure alignment with treatment plans. Ongoing monitoring of patient-reported app data for accuracy and clinical relevance is required.

PR-2. What are obstacles that prevent development, deployment, or effective utilization of the most useful and innovative applications for physician workflows, such as quality measurement reporting, clinical documentation, and billing tasks? How could these obstacles be mitigated?

Fragmented EHR architectures inhibit integration, with 63% of providers reporting incompatibility between new apps and legacy systems. Lack of standardized clinical terminologies forces manual mapping of lab results and vital signs across platforms. CMS could mandate FHIR-based app certification and fund open-source toolkits for small practices.

More importantly for smaller and rural providers, the primary obstacle is cost. Small and rural healthcare providers in the U.S. face significant financial barriers to adopting health IT systems. Upfront costs for certified EHR software often exceed $15,000–$50,000 per provider,

compounded by infrastructure upgrades such as reliable broadband (absent in 25% of rural hospitals) and modern hardware, which can total $100,000–$500,000. Ongoing expenses—including 15–20% annual maintenance fees, $5,000–$20,000/month for legacy-to-FHIR middleware, and staff training—strain limited budgets. These costs are 5–7x higher relative to revenue compared to urban peers, forcing many rural clinics to delay upgrades or rely on outdated systems.

Interoperability and workforce gaps exacerbate the challenge. Participation in data-sharing networks such as TEFCA requires $50,000–$250,000 annually in fees, while customizing FHIR APIs for rural workflows adds $50,000–$150,000. Rural areas also face IT specialist shortages, requiring 30–50% salary premiums to attract contractors. These technical and staffing hurdles create fragmented systems that struggle to share data, worsening care coordination and efficiency.

Policy shortfalls and uncertain returns further deter investment. Value-based care savings (e.g., reduced hospitalizations) take 3–5 years to materialize, clashing with fee-for-service revenue priorities. While federal programs such as the FCC's Telehealth Fund offset only 15–20% of costs, rural providers remain caught between rising tech demands and stagnant reimbursement models. This mismatch perpetuates care disparities, as outdated systems limit access to innovations such as AI-driven diagnostics or real-time population health tools available in urban centers.

PR-3. How important is it for healthcare delivery and interoperability in urban and rural areas that all data in an EHR system be accessible for exchange, regardless of storage format (for example, scanned documents, faxed records, lab results, free text notes, structured data fields)? Please address all of the following:

a. Current challenges in accessing different data formats.

b. Impact on patient care quality.

c. Technical barriers to full data accessibility.

d. Cost or privacy implications of making all data formats interoperable.

e. Priority level compared to other interoperability needs.

Ensuring all data within EHR systems, whether structured fields, scanned documents, faxed records, or free-text notes, is accessible for exchange is critical for equitable healthcare delivery and interoperability across urban and rural U.S. regions. Rural providers, who often rely on faxes and scanned records due to legacy systems and fragmented infrastructure, face severe care coordination gaps when these formats cannot be integrated into modern FHIR-based exchanges, leading to incomplete patient histories, duplicated tests, and preventable errors (e.g., 18% of adverse drug events stem from missing records). Urban health systems, while better resourced, still grapple with siloed data; imaging studies or specialist notes trapped in PDFs often delay treatment for transferred patients. Standardizing access to all data formats would bridge disparities, enabling AI-driven tools to parse unstructured notes for risk stratification or SDOH insights, while reducing administrative bottlenecks. However, technical and financial barriers such as OCR conversion costs for scanned documents ($5–$20 per patient) and legacy EHR upgrades disproportionately burden rural providers, exacerbating care delays. Full data accessibility is thus both a clinical necessity and a structural imperative to achieve seamless, patient-centered care, though it requires targeted funding and policy support to address rural-urban resource divides.

PR-4. What changes or improvements to standards or policies might be needed for patients' third-party digital products to have access to administrative workflows, such as auto-populating intake forms, viewing provider information and schedules, and making and modifying an appointment?

CMS should mandate SMART on FHIR v4.0 for auto-populating intake forms using USCDI v3 elements. Provider directory APIs must include real-time scheduling slots using FHIR Schedule resources. Policy changes should prohibit EHR vendors from charging fees for third-party app connections.

CMS should also fund continued work on universal provider and patient appointing standards to make it easier for providers to share appointment availability with patients and PHRs and for patients to look for appointments **across** providers' schedules. While HL7 has established mature messaging standards (SIU in V2.x) and a modern FHIR Appointment resource for appointment scheduling, a single, universal API for provider appointment scheduling does not yet fully exist. Further, ONC is actively advancing API certification and interoperability, which could support universal appointment APIs, but has not yet mandated a specific appointment scheduling API standard. Industry adoption of HL7 SIU and FHIR Appointment varies, and full universality (i.e., a single API standard used by all systems) remains a work in progress.

In summary, while both HL7 and the ONC have made substantial progress in standardizing appointment data exchange, a truly universal provider appointment API and standard is still evolving, with FHIR Appointment and ONC's API certification efforts representing the most promising foundation for future development. Most importantly, from a patient perspective in the United States, it is still nearly impossible to look for urgent or primary care appointments from a PHR across providers in any meaningful way. This is what patients want and need.

PA-2. How can CMS encourage payers to accelerate the implementation and utilization of APIs for patients, providers, and other payers, similar to the Blue Button 2.0 and Data at the Point of Care APIs released by CMS?

CMS should provide stronger incentives, tied to payor Star Ratings, for payers to adopt and promote APIs for data access between government entities (CMS, DoD, VA, etc.), providers and patients. Accelerating API implementation for payers, providers, and patients is critical to advancing interoperability, reducing administrative burdens, and empowering data-driven care. By incentivizing adoption of standards such as Blue Button 2.0 (which uses FHIR-based APIs to give patients secure access to Medicare claims data) and the Data at the Point of Care API (which enables provider access to patient information during clinical workflows), CMS fosters seamless data exchange that improves care coordination and clinical decision-making. These APIs reduce redundant processes, such as manual prior authorization workflows, while enhancing transparency for patients and providers. For payers, standardized APIs future-proof infrastructure for emerging value-based care models and regulatory requirements such as CMS-0057-F, which mandates payer-to-payer data sharing to maintain care continuity. By aligning incentives, CMS ensures broader participation in a connected ecosystem where real-time data access becomes the norm, driving innovation in patient-facing tools and reducing systemic inefficiencies.

- Tie API adoption to Star Ratings: Offer +0.5-star boosts for plans achieving 90% FHIR-based prior auth automation.

- Expand the DPC model: Require payers to implement Bulk FHIR exports for HEDIS reporting by 2026, mirroring NCQA's coalition.
- Launch FHIR certification programs: Fund developer training on CARIN Blue Button standards to reduce integration costs.

PA-4. What would be the value to payers of a nationwide provider directory that included FHIR end points and used digital identity credentials?

CMS should fund, promote and run a nationwide provider directory and incentivize providers and payors to adopt and support these initiatives. A U.S. nationwide provider directory run by CMS is essential because the current healthcare directory landscape is fragmented, causing significant challenges for patients, providers, and payers. Patients often struggle to find accurate, up-to-date information about in-network providers who meet their specific needs, while providers face redundant and burdensome reporting requirements to multiple databases, resulting in widespread inaccuracies and inefficiencies. A centralized, CMS-managed directory would serve as a trusted, authoritative resource, improving access to care, supporting patient choice, and enabling easier comparison of health plan networks. By leveraging modern interoperable technology and streamlining information submission, such a directory would facilitate care coordination, enhance health information exchange, and reduce administrative costs across the industry, ultimately supporting better health outcomes and more efficient public health reporting.

A nationwide provider directory integrating FHIR endpoints and digital identity credentials would offer payers transformative operational, financial, and strategic benefits. By centralizing validated provider data (including specialties, affiliations, and FHIR API endpoints), payers could reduce administrative costs by up to $2 billion annually through eliminating redundant credentialing processes and manual updates across fragmented systems. Digital identity verification would minimize fraud risks by ensuring only authenticated providers are included, reducing claim denials and improper payments linked to invalid credentials.

FHIR-enabled endpoints would accelerate interoperability, allowing payers to automate prior authorization, referrals, and care coordination by programmatically accessing real-time provider availability and services. This aligns with CMS mandates for public FHIR-based provider directories while enabling payers to compete on network quality. Prospective members could compare in-network specialists, languages spoken, or telehealth capabilities via third-party apps.

Additionally, verified digital identities would streamline payer-to-payer data exchanges required under CMS-0057-F, ensuring continuity when switching plans. The directory's FHIR infrastructure also positions payers to monetize data partnerships with digital health innovators seeking API access to provider networks. Real-time credential monitoring would ensure compliance with state licensing and federal regulations, avoiding penalties while maintaining network adequacy.

TD-1. What short-term (next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?

To stimulate developer interest in building transformative digital health tools for Medicare beneficiaries and caregivers, CMS must leverage its regulatory authority and purchasing power to create a standardized, government-backed ecosystem that prioritizes patient needs over vendor profits. In the short term (0-2 years), CMS should focus on dismantling existing barriers to data access and interoperability while providing developers with the infrastructure and incentives to innovate. Building on the Blue Button 2.0 API and Data at the Point of Care program, CMS could mandate that all Medicare-participating providers and payers expose FHIR R4.0.1 endpoints for clinical, claims, and social determinants of health (SDOH) data. This would give developers real-time access to the full spectrum of patient information needed to build care coordination tools, reducing the current burden of integrating with dozens of proprietary EHR systems. To further lower development costs, CMS should launch a National Digital Health Sandbox, providing synthetic Medicare datasets and pre-certified API connectors that mimic real-world TEFCA QHIN exchanges. This sandbox would allow developers to test apps against simulated scenarios—such as managing a diabetic patient's transition from hospital to home care—without navigating complex data use agreements.

Simultaneously, CMS must address the "last mile" adoption challenge by tying Medicare reimbursement to digital health integration. For example, hospitals achieving >80% patient enrollment in certified care navigation apps could receive a 2% boost in Value-Based Purchasing Program scores, while primary care providers using CMS-validated SDOH screening tools might qualify for enhanced Chronic Care Management payments. To ensure equity, CMS should subsidize broadband and device access for low-income beneficiaries through Medicaid IT modernization funds, creating a guaranteed user base that attracts developer investment.

For longer-term (2+ years) transformation, CMS should evolve TEFCA into a National Health Data Utility, mandating that all QHINs support bidirectional data flows between patients, providers, and apps. This utility would serve as a public infrastructure layer, akin to the electrical grid, where developers can plug into standardized APIs to access de-identified population health data for AI training, while patients retain granular control over their information via NIST IAL2 digital credentials. CMS could further catalyze innovation by requiring EHR vendors to embed government-developed app stores within their platforms, featuring vetted tools that meet stringent accessibility, security, and interoperability criteria. These app stores would function similarly to the VA's Lighthouse program, but with CMS enforcing a "no proprietary fees" rule to prevent vendor lock-in.

To sustain developer engagement, CMS should establish a Digital Health Evidence Accelerator Program, funding pilot projects that demonstrate measurable improvements in outcomes such as 30-day readmissions or medication adherence. Successful pilots would receive fast-tracked certification and automatic inclusion in Medicare Advantage supplemental benefit packages, creating a clear pathway to scale. Finally, CMS must overhaul its certification processes to prioritize API-first design, requiring all new health IT systems to expose 100% of patient data elements via FHIR—including legacy formats such as faxed records through AI-driven OCR-to-FHIR conversion tools.

Finally, CMS must recognize that proprietary system vendors such as Epic and Cerner have little incentive to actively support small vendor, niche patient facing use cases or functionality. Their API pricing models and developer on boarding requirements are tailored for large institution or for profit commercial business models. CMS should require that patient data access services, non-profits, and other social benefit institutions creating solutions for individual patients receive free or significantly discounted API and developer access. If Government policy and incentives remain disproportionately focused on commercial interests, CMS will miss an opportunity to radically transform the national health IT infrastructure into an engine of the patient-centered digital health revolution.

TD-3. Regarding digital identity implementation:

a. What are the challenges and benefits?

b. How would requiring digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 CSPs) impact cybersecurity and data exchange?

c. What impact would mandatory use of the OpenID Connect identity protocol have?

The implementation of robust digital identity systems in healthcare, driven by CMS and federal agencies through frameworks such as TEFCA and QHINs, is critical to addressing systemic fragmentation and security vulnerabilities. By mandating standardized, government-backed solutions, CMS can overcome proprietary barriers and ensure equitable, secure access for all patients.

Challenges surrounding digital identity implementations stem from fragmented systems and misaligned incentives. Patient matching errors persist due to reliance on error-prone demographic data, with legacy EHRs often lacking interoperability. For example, 45% of provider directories inaccuracies hinder care coordination. Additionally, resistance from vendors prioritizing proprietary systems over open standards creates silos, while marginalized populations such as rural or elderly patients face accessibility gaps due to limited digital literacy or broadband access.

Benefits are transformative. A unified digital identity framework would eliminate redundant logins, reduce fraud (estimated at $6.3B annually in Medicare), and enhance care continuity. For example, NIST IAL2/AAL2 credentials, as adopted by CLEAR and Login.gov, provide cryptographic assurance of patient identity, ensuring that critical data such as advance directives or medication lists are accurately linked to individuals. This reduces medical errors and streamlines emergencies, where instant access to verified records can save lives.

Requiring NIST 800-63-3 IAL2/AAL2 credentials (e.g., Login.gov, CLEAR) would revolutionize cybersecurity and data exchange. These credentials, validated through rigorous processes such as Kantara Initiative audits, ensure that only authenticated users access sensitive data, mitigating breaches and identity theft. For example, CMS's planned federated identity solution would let patients use a single credential across providers, payers, and apps, eliminating the need to manage multiple passwords. This interoperability is vital for value-based care, where accurate attribution of outcomes relies on seamless data flows.

However, equity must be prioritized. Mandates should include offline access options, such as USB-C health keys at rural kiosks, and subsidies for low-income beneficiaries to obtain devices.

Without these measures, credential requirements could exclude vulnerable populations, exacerbating disparities.

Adopting OpenID Connect as the federal authentication standard would harmonize access across healthcare's fractured landscape. This protocol, already proven in sectors such as banking, enables single sign-on (SSO) and granular consent management. Integrating OpenID Connect with FHIR APIs allows patients to authorize apps to access specific data (e.g., lab results but not mental health records), aligning with HIPAA's minimum necessary standard.

CMS can leverage OpenID Connect to enforce patient-directed data sharing, as seen in Norway's health network, where users control access via digital wallets. This would empower patients while reducing administrative burdens; providers could pull verified records instantly instead of manually reconciling duplicates. However, legacy EHRs using HL7v2 or SAML require federal support to transition, such as CMS-funded FHIR gateway services.

CMS can play a strategic role in national standardization effort, both as architect and enforcer. Through its policies, technical leadership, and financial incentives, it must:

- Mandate Participation: Condition Medicare reimbursement on TEFCA/QHIN compliance, requiring EHR vendors to expose OpenID Connect endpoints and NIST-aligned credentials.
- Fund Transition Infrastructure: Subsidize legacy system upgrades and rural kiosks, ensuring no patient is left behind.
- Certify and Monitor: Launch a Federal Identity Assurance Program to audit vendors' adherence to OpenID Connect and NIST standards, with penalties for non-compliance.

By centralizing governance and rejecting the vendor-driven status quo, CMS can transform digital identity from a commercial battleground into a public utility, securing data, empowering patients, and enabling the promise of interoperable care.

TD-13. What new opportunities and advancements could emerge with APIs providing access to the entirety of a patient's electronic health information (EHI)?
a. What are the primary obstacles to this?
b. What are the primary tradeoffs between USCDI and full EHI, especially given more flexible data processing capabilities today?

The ability for a healthcare provider and the individual patient themselves to access a patient's entire EHI via standardized APIs represents a transformative leap toward patient-centered care and innovation. By mandating such access through frameworks such as TEFCA and QHINs, CMS can unlock unprecedented opportunities while dismantling the systemic barriers that currently fragment care and stifle progress.

With comprehensive EHI APIs, patients and providers could harness longitudinal health narratives that integrate structured data such lab results with unstructured elements such as clinician notes, imaging annotations, and social determinants of health (SDOH) documented in free text. For example, a patient with diabetes could use an app to correlate decade-old HbA1c trends with recent ophthalmology notes, identifying early retinopathy risks that discrete data points alone might miss. Researchers could analyze full EHI to uncover patterns in rare diseases, while AI models trained on rich, contextual data could predict sepsis or opioid dependency with far greater accuracy. CMS

could further catalyze public health by embedding real-time EHI streams into TEFCA's backbone, enabling rapid detection of disease outbreaks or environmental health threats through aggregated, de-identified data.

The primary obstacles to full health data utilization are deeply entrenched in healthcare's technical and economic fabric. Legacy EHR systems still reliant on HL7v2 and proprietary formats cannot natively support modern FHIR APIs for unstructured data. Even when APIs exist, vendors often impose arbitrary limits, such as throttling query speeds or omitting "non-core" data such as imaging files, to maintain control over data ecosystems. Privacy concerns, while valid, are exacerbated by inconsistent consent management tools and patient-matching errors, which persist due to the lack of a national identity resolution strategy. Perhaps most critically, vendor resistance to open data sharing undermines progress; EHR developers profit from siloed systems that lock providers into proprietary platforms, creating disincentives to prioritize interoperability.

The US Core Data for Interoperability (USCDI) provides a critical but limited foundation, standardizing elements such as medications and allergies to ensure baseline consistency. However, its structured nature excludes the nuanced, unstructured data that often drive clinical decisions—such as a provider's rationale for deviating from a treatment guideline or a caregiver's observations about a patient's home environment. While USCDI ensures interoperability "out of the box," it sacrifices the depth needed for personalized care and advanced analytics. Modern technologies such as natural language processing (NLP) can parse free-text notes and imaging reports, but current policies do not require EHRs to expose this data via APIs. The tradeoff, then, is between standardized simplicity (USCDI) and contextual richness (full EHI). CMS can bridge this gap by expanding USCDI to include "USCDI+", a modular framework that mandates FHIR-based access to unstructured data while preserving backward compatibility. For example, a cardiology app could access structured echocardiogram metrics via USCDI while using NLP to extract ejection fraction notes from PDFs, all through a single API.

To realize this vision, CMS must act as both architect and enforcer, where its policies and incentives:

- Mandate Full EHI APIs via TEFCA: Require all QHIN participants to expose 100% of patient data—structured and unstructured—through FHIR R4.0.1, with penalties for non-compliance.
- Modernize Legacy Systems: Fund EHR vendors to replace HL7v2 interfaces with FHIR gateways, prioritizing rural and safety-net providers.
- Standardize Unstructured Data: Develop NLP-powered tools to convert scanned documents, faxes, and free-text notes into FHIR DocumentReference resources, ensuring they flow seamlessly into APIs.
- Enforce Patient Control: Deploy NIST IAL2 digital identity credentials to let patients granularly share data (e.g., "only my oncology notes") while auditing vendor compliance with real-time transparency tools.

By treating EHI as a public good rather than a proprietary asset, CMS can eliminate the paywalls and fragmentation that currently limit innovation. This approach aligns with the 21st Century Cures Act's intent while ensuring that the benefits of interoperability—personalized care,

equitable access, and scientific discovery—are universally accessible, not just commercially exploitable.

VB-1. What incentives could encourage APMs such as accountable care organizations (ACOs) or participants in Medicare Shared Savings Program (MSSP) to leverage digital health management and care navigation products more often and more effectively with their patients? What are the current obstacles preventing broader digital product adoption for patients in ACOs?

Cognitive sees firsthand the crossroads facing ACOs and participants in the Medicare Shared Savings Program (MSSP) as they strive to deliver on the promise of value-based care. Digital health management and care navigation products have the potential to revolutionize how care is delivered, coordinated, and experienced by patients. Yet, despite their promise, these tools remain underutilized. The reasons are complex, but the opportunities to accelerate adoption are clear and within reach.

To encourage ACOs and similar organizations to embrace digital health more fully, a thoughtful mix of incentives is essential. Financial incentives especially remain a powerful lever. For example, tying bonuses and enhanced shared savings directly to measurable digital milestones, such as achieving high rates of FHIR API integration or demonstrating reductions in hospitalizations through AI-driven risk stratification, can motivate organizations to prioritize digital transformation. We also see value in rewarding both providers and patients for engaging with digital tools. A dual-approach system where patients receive small rewards for participating in telehealth visits or using remote monitoring devices and providers are recognized for driving digital engagement not only fosters adoption but creates a sense of shared purpose.

Incentives need not be purely financial; recognition programs that celebrate digital health champions within organizations can help build a culture of innovation and continuous improvement. Public acknowledgment, digital leader certifications, and peer-led learning collaboratives can all play a role in making digital adoption something that teams aspire to, not just comply with.

Despite these potential motivators, significant obstacles remain. Many ACOs, especially smaller or physician-led ones, face real financial and resource constraints when it comes to investing in new technology. Legacy EHR systems and fragmented data sources make integration of digital tools challenging, often forcing clinicians into inefficient manual workflows. Inaccurate provider directories further complicate care coordination, leading to administrative headaches and increased costs. On top of these technical challenges, there's the human element: clinicians and staff may be resistant to change, lack adequate training, or feel overwhelmed by the learning curve associated with new digital platforms.

Patients also face barriers. Poorly designed interfaces, lack of digital literacy, and limited access to devices or reliable internet can all limit engagement, particularly among older adults or those in underserved or rural communities. Regulatory and reimbursement constraints, along with concerns about privacy and liability, add further complexity to the adoption landscape.

Addressing these challenges requires a comprehensive approach. Federal and state support, such as grants or low-interest loans, can help ACOs modernize legacy systems and invest in digital

infrastructure. Mandating FHIR-based provider directories and data standards would make significant strides toward solving interoperability issues. At the same time, robust education and change management programs are needed to build digital literacy and confidence among clinicians, staff, and patients alike. Co-designing digital tools with input from users ensures that solutions are accessible, intuitive, and relevant to the populations they serve.

Incremental, modular implementation strategies, rather than wholesale, overnight transformations can be risk-averse solutions. Organizations can start with high-impact use cases, such as remote monitoring for high-risk patients or digital care navigation for those with complex needs, and expand as they gain experience and confidence. Continuous feedback and iterative improvement are key to sustaining momentum and maximizing the benefits of digital adoption.

Ultimately, digital health management and care navigation products are not just nice-to-have add-ons; they are foundational to the future of value-based care. By aligning incentives, addressing both technical and human barriers, and keeping patient and provider experience at the center, ACOs and MSSP participants can move from isolated digital experiments to integrated, effective care delivery.

VB-2. How can key themes and technologies such as artificial intelligence, population health analytics, risk stratification, care coordination, usability, quality measurement, and patient engagement be better integrated into APM requirements?

The true promise of value-based care lies in seamlessly weaving advanced technologies such as AI, population health analytics, risk stratification, care coordination, usability, quality measurement, and patient engagement into the very fabric of Alternative Payment Model (APM) requirements. Currently, many organizations treat these capabilities as checkboxes rather than essential tools for transforming care. APMs must move beyond compliance and instead require that these technologies are operationalized in daily workflows, directly supporting better outcomes for patients and providers alike.

One powerful way to do this is by mandating real-time, interoperable data exchange. For example, requiring FHIR-based event notifications, aligned with the latest national endpoint directories, would allow care teams to coordinate instantly when a patient is admitted or discharged. Such real-time communication has already shown the potential to reduce preventable readmissions by a third in CMS pilot programs, demonstrating the tangible value of embedding technology into care delivery.

But integration must go deeper. To truly enable AI and population health analytics, APMs should require the use of advanced data standards such as USCDI v3 and beyond. These standards now include a broader range of data elements, such as SDOH and genomic information, which are critical for building accurate risk stratification models. By incorporating these data types into quality reporting, APMs can ensure that risk models account for factors such as food insecurity or genetic predispositions, variables that are still missing from the majority of current contracts. This richer data environment allows AI to not only identify high-risk patients more precisely but also to recommend timely, personalized interventions that can prevent complications and reduce costs.

Care coordination and patient engagement are also ripe for deeper integration. AI-powered tools can automate outreach, provide tailored health reminders, and facilitate two-way communication

between patients and care teams, making it easier for patients to stay engaged in their own care. By requiring that APM participants deploy these tools and measure their impact through patient engagement metrics, APMs can ensure that technology is used to close gaps in care, rather than widen them.

Usability is another critical theme. Too often, digital tools are implemented without enough attention to the end-user experience, leading to frustration and low adoption. APM requirements should include usability standards and require regular feedback from both clinicians and patients to drive continuous improvement. This not only boosts adoption but also ensures that technology serves as a help, not a hindrance, to care delivery.

Finally, quality measurement itself should evolve to reflect the new digital reality. Rather than relying solely on retrospective claims data, APMs can require real-time quality measurement using data from EHRs, remote monitoring devices, and patient-reported outcomes. This enables more responsive, proactive management of patient populations and allows organizations to adjust their strategies quickly when gaps are identified.

By embedding these advanced technologies and themes directly into APM requirements through mandates, incentives, and ongoing measurement, we can move from theoretical potential to real-world impact, ensuring that every care decision is informed by holistic, actionable insights and that the promise of value-based care is fully realized for every patient.

VB-3. What are essential health IT capabilities for value-based care arrangements?

a. Examples (not comprehensive) may include: care planning, patient event notification, data extraction/normalization, quality performance measurement, access to claims data, attribution and patient ID matching, remote device interoperability, or other patient empowerment tools.

b. What other health IT capabilities have proven valuable to succeeding in value-based care arrangements?

The foundation of successful value-based care (VBC) is not just the presence of health IT tools, but their seamless integration and interoperability across the care continuum. Essential health IT capabilities for VBC arrangements must empower providers to deliver coordinated, data-driven, and patient-centered care—transforming isolated functionalities into a connected ecosystem that supports both clinical and financial goals.

A core capability is robust, FHIR-enabled care planning. Modern care planning modules should do more than document interventions; they should automatically incorporate key SDOH insights, such as housing instability or transportation barriers. Medicaid innovation projects have shown that addressing these factors through integrated care planning can close up to 40% of care gaps in underserved populations, demonstrating the tangible impact of embedding SDOH into the workflow.

Real-time patient event notifications are another critical capability. Leveraging FHIR Subscriptions, care teams can receive instant alerts about significant clinical events such as hospital admissions, abnormal lab results, or care plan changes, enabling rapid response and better care coordination. Systems such as these have already improved patient outcomes and safety by ensuring that all relevant stakeholders are kept up to date and can act quickly when needed.

Data extraction, normalization, and integration are also essential. Effective value-based care depends on the ability to pull structured and unstructured data from multiple EHRs, clean and standardize it using AI-driven tools and natural language processing and map it to industry standards such as FHIR and HL7. This ensures a unified, high-quality patient record that supports accurate reporting, risk adjustment, and regulatory compliance. Bulk FHIR exports, for example, have already reduced manual data abstraction by 40% for MSSP participants, freeing clinicians to focus more on patient care.

Access to both clinical and claims data is vital for a complete picture of patient care. Integrating these datasets allows providers to close gaps in care, avoid medication errors, and support evidence-based treatment decisions. Health plans and providers that partner to share requirements and clinical data in near real-time are better equipped to coordinate care for vulnerable populations and meet increasing quality reporting requirements.

Provider directories, often overlooked, are foundational for accurate referrals and network management. NDH-compliant directories, maintained with verified credentials and regularly updated via API, help eliminate the 45% inaccuracy rate still seen in many directories, reducing referral errors and compliance risks.

Transparency and fairness in AI-driven tools are increasingly important as these technologies become core to risk stratification, decision support, and quality measurement. The new HTI-1 rule from HHS requires algorithm transparency, ensuring clinicians have access to information about the fairness, validity, and safety of the AI tools they use. This is crucial for avoiding biases that could worsen health disparities and for building trust in advanced analytics.

Other essential capabilities include patient empowerment tools such as electronic portals and mobile apps for accessing health records, telehealth for expanding access and ongoing monitoring, and remote device interoperability for integrating data from wearables and home monitoring devices directly into care management workflows. Attribution and patient ID matching, often supported by advanced data management systems, ensure that every patient is accurately tracked across providers and settings, which is fundamental for both care quality and financial accountability.

Ultimately, interoperability is the thread that ties all these capabilities together. By maximizing interoperability through FHIR, standardized data models, and open APIs organizations can optimize care efficiency, minimize redundant tests, and create a holistic, patient-centered view that is essential for achieving the goals of value-based care. Healthcare IT is not just a set of tools, but the scaffolding for care delivery that is as precise as it is compassionate.

VB-4. What are the essential data types needed for successful participation in value- based care arrangements?

For VBC arrangements to thrive, the completeness and quality of data are absolutely essential. Data is not just as a technical requirement, but as the foundation for understanding the whole patient and delivering truly effective, equitable care. The most successful VBC organizations are those that can harness a comprehensive set of data types, integrate them seamlessly, and use them to inform every aspect of care delivery and population health management.

First and foremost, claims data is indispensable. Real-time access to adjudicated claims, such as through expanded Blue Button 2.0 APIs, enables organizations to monitor utilization, track costs, and anticipate billing issues before they escalate. This kind of timely claims data has already helped reduce administrative costs by up to 25% in Medicare Advantage plans, and it empowers APMs to optimize resource allocation and financial performance.

Equally critical is the integration of EHRs, which provide the clinical backbone of patient care. EHRs capture diagnoses, lab results, medications, procedures, and encounter histories, offering a longitudinal view of each patient's health journey. However, the value of EHR data is amplified when it is standardized and interoperable, using frameworks such as USCDI v3 or higher. These standards ensure that data elements such as health insurance information, functional and mental health status, and even genomic data are consistently captured and shared across systems.

SDOH data are increasingly recognized as essential for VBC success. Factors such as food insecurity, housing instability, transportation access, and social support can profoundly influence health outcomes and care costs. Yet, only about 30% of EHRs currently capture SDOH codes such as food insecurity, leaving many care plans blind to these root causes of poor outcomes. Mandating structured SDOH documentation for all ACO-attributed lives, using USCDI v3+ elements, would enable interventions as impactful as the VA's housing-first initiatives, which have reduced emergency department visits by 22% in high-risk cohorts.

Other key data types include patient demographics, risk adjustment scores such as HCC and RAF, and health risk assessments, all of which are vital for accurate patient attribution, risk stratification, and performance measurement. Admission, discharge, and transfer (ADT) notifications, as well as data from HIEs, provide real-time insights into patient movement and care transitions—critical for proactive care coordination.

Patient-generated health data, such as information from remote monitoring devices or patient portals, further enrich the clinical picture and support ongoing engagement and self-management. Finally, quality performance metrics such as HEDIS measures are necessary for tracking outcomes, benchmarking performance, and meeting reporting requirements.

For VBC to reach its full potential, all of these data types must be accessible, timely, accurate, and actionable. They should be shared across an interoperable ecosystem, governed by strong data standards, and leveraged by advanced analytics to drive continuous improvement.

VB-5. In your experience, how do current certification criteria and standards incorporated into the ONC Health IT Certification Program support value-based care delivery?

VB-6. What specific health information technology capabilities that could benefit APMs are not currently addressed by existing certification criteria and standards that should be included under the ONC Health IT Certification Program?

The ONC Health IT Certification Program has laid a strong foundation for value-based care (VBC) by driving interoperability, data access, and transparency. The program's certification criteria, such as FHIR API requirements and adoption of the US Core Data for Interoperability (USCDI) v3, have been instrumental in enabling automated processes such as prior authorization, which has helped reduce processing delays by a third in TEFCA-aligned health systems. These requirements

also support secure, standards-based data exchange, which is crucial for VBC providers who rely on timely, comprehensive information to coordinate care and measure outcomes.

The recent HTI-1 rule marks another important step by establishing transparency requirements for AI and predictive algorithms used in certified health IT. This means that clinicians will have access to baseline information about the algorithms they use, including their fairness, validity, and safety; these are vital safeguards as AI becomes more embedded in risk stratification, clinical decision support, and population health analytics. The rule also mandates the use of USCDI v3, which expands the types of data available for care coordination and equity-focused interventions, and encourages secure, standards-based exchange under TEFCA.

However, as VBC evolves, so must certification standards. In our experience, current criteria don't yet fully address the dynamic needs of APMs and value-based organizations. For example, while FHIR APIs and bulk data access are now part of certification, there is still a gap around nationwide provider directory standards such as NDH endpoint validation, which are essential for reducing referral errors and improving network management. Certification should also evolve to include more robust support for cross-population analytics, allowing organizations to efficiently analyze data at scale for quality measurement, risk adjustment, and performance improvement.

Another critical area is algorithm transparency and fairness. While the HTI-1 rule introduces baseline requirements, certification should go further by mandating regular validation and bias testing for AI models, especially those used in rural or underserved populations where misclassification can exacerbate disparities. This would help prevent the kinds of "black box" algorithms that have been shown to mis-rank up to 15% of rural patients in risk scores.

Additionally, certification criteria should expand to include capabilities that directly support patient empowerment and engagement, such as advanced care planning tools, real-time event notifications, and remote device interoperability. These features are becoming essential for VBC success but are not yet comprehensively addressed in current standards.

Ultimately, ONC certification should serve as a launchpad, not a ceiling, for health IT innovation. By continually updating certification criteria to reflect emerging needs including advanced analytics, AI transparency, provider directory accuracy, and patient engagement, ONC can ensure that certified health IT is not only compliant, but also truly supportive of equitable, high-quality, value-based care.

VB-7. How can technology requirements for APMs, established through CEHRT or other pathways, reduce complexity while preserving necessary flexibility?

VB-8. How can other HHS policies supplement CEHRT requirements to better optimize the use of digital health products in APMs? As an example, requirements under the Conditions of Participation for hospitals (42 CFR 482.24(d)) require hospitals to transmit electronic patient event notifications to community providers. What barriers are in place preventing APM participants from receiving the same notifications?

The promise of value-based care is often hampered by the complexity and fragmentation of technology requirements for APMs. Certified EHR Technology (CEHRT) and related standards are vital for ensuring interoperability and data quality, but the current landscape is crowded with redundant and sometimes conflicting requirements. This complexity can overwhelm APM

participants, especially when trying to balance compliance with the flexibility needed to innovate and serve diverse patient populations.

One of the most effective ways to reduce this complexity is by consolidating standards and aligning certification pathways. For example, adopting unified models such as the FAST Implementation Guide's endpoint directory could replace a dozen or more legacy data formats, much like the Department of Veterans Affairs successfully consolidated 22 referral networks into a single FHIR-based system. Such consolidation streamlines workflows, reduces technical burden, and makes it easier for APMs to connect with a broad array of partners and services.

Further, aligning CEHRT requirements with TEFCA for QHINs would allow APMs to submit data through a single source for multiple regulatory needs. This alignment has already enabled some ACOs to achieve nearly universal compliance with reporting requirements, while eliminating the need for duplicative submissions and audits. By making CEHRT the backbone for both clinical and administrative data exchange, APMs can focus resources on patient care rather than on navigating a maze of technical standards.

However, flexibility must be preserved. Healthcare organizations vary widely in their size, resources, and patient populations. Technology requirements should set clear interoperability and data quality baselines, while allowing room for organizations to choose solutions that fit their unique workflows and care models. This approach encourages innovation and lets APMs tailor their digital strategies to the needs of their communities, rather than forcing a one-size-fits-all solution.

Other HHS policies can play a crucial role in supplementing CEHRT requirements and optimizing digital health adoption in APMs. For example, while hospitals are required under the Conditions of Participation (42 CFR 482.24(d)) to transmit electronic patient event notifications to community providers, APM participants often lack similar mandates or technical infrastructure to receive these notifications. This creates a critical gap in care coordination, especially during transitions between hospital and outpatient settings.

Extending FHIR-based alert rules through TEFCA could help close this gap, ensuring that APMs receive timely, actionable notifications about patient events. Technical barriers remain, however, such as the absence of standardized FHIR Schedule resources, which currently disrupts up to 30% of care transitions by making it difficult to coordinate follow-up appointments and services. Addressing these gaps will require both regulatory action and industry collaboration to ensure that all APM participants, regardless of their technical maturity, can benefit from real-time care coordination tools.

In summary, reducing complexity in APM technology requirements is best achieved by consolidating and aligning standards, leveraging CEHRT as a single source of truth, and supplementing these requirements with targeted HHS policies that address gaps in interoperability and care coordination. At the same time, preserving flexibility ensures that organizations can adapt and innovate to meet the evolving needs of value-based care.

VB-11. What specific interoperability challenges have you encountered in implementing value-based care programs?

VB-12. What technology standardization would preserve program-specific flexibility while promoting innovation in APM technology implementation?

VB-13. What improvements to existing criteria and standards would better support value-based care capabilities while reducing provider burden?

VB-14. How could implementing digital identity credentials improve value-based care delivery and outcomes?

VB-15. How could a nationwide provider directory of FHIR endpoints help improve access to patient data and understanding of claims data sources? What key data elements would be necessary in a nationwide FHIR endpoints directory to maximize its effectiveness?

Implementing VBC programs has brought to light a host of interoperability challenges, many of which stem from the fragmented nature of healthcare data and the lack of a unified approach to digital identity and endpoint management. These issues can undermine the promise of VBC, both technically and operationally.

One of the most persistent challenges is the fragmentation of data across multiple, often incompatible systems. Providers, payers, and patients each interact with a patchwork of EHRs, claims systems, and third-party applications, all using different identifiers, credentials, and endpoint definitions. This fragmentation leads to incomplete longitudinal records, blind spots in care, and increased administrative burden for clinicians who must navigate disparate systems to piece together a patient's full story. As one of our engineers astutely noted, mapping a provider to one or more endpoints is not just a directory challenge it's fundamentally an identity challenge. Without a unified, persistent digital identity for every party in the healthcare ecosystem including providers, organizations, devices, and patients, true interoperability and data completeness remain out of reach.

The coexistence of legacy systems with modern IT environments further complicates matters, as many older platforms lack the standardized protocols and instrumentation necessary for seamless integration. This technical gap makes it difficult to gain a holistic view of application and patient data, hindering both performance monitoring and care coordination. Additionally, resistance to change and the administrative burden of adapting to new workflows can slow technology adoption and increase provider fatigue.

To address these challenges, technology standardization must strike a careful balance between providing a strong, consistent foundation and allowing for program-specific flexibility and innovation. Modular, scalable frameworks, such as the FHIR standard, enable organizations to implement only the resources and profiles relevant to their needs while maintaining a common language for data exchange. This approach allows for customization where necessary, without sacrificing the integrity or security of the overall system. Regular feedback loops and iterative updates ensure that standards evolve alongside emerging technologies and user needs.

Improvements to existing criteria and standards should focus on reducing provider burden by automating data validation, streamlining directory updates, and supporting bulk data operations. For example, a centralized, validated national directory of FHIR endpoints would eliminate the

need for providers to repeatedly submit the same information to multiple stakeholders, saving the industry significant time and resources. Enhanced standards for digital identity credentials would further reduce administrative overhead by enabling secure, automated verification of provider and patient identities across systems.

Implementing robust digital identity credentials is a foundational step toward improving VBC delivery and outcomes. With secure, persistent digital identities, organizations can confidently share sensitive health information, coordinate care across settings, and enable patients to access their own records without fear of unauthorized disclosure. Digital identity also supports better fraud prevention, streamlined onboarding, and more efficient patient engagement, ultimately leading to improved care quality and reduced costs.

A nationwide provider directory of FHIR endpoints would be transformative for access to patient data and understanding claims data sources. Such a directory would allow authorized users, such as providers, payers, and patients, to quickly locate the correct endpoint for any provider or organization, facilitating seamless data exchange and reducing errors in referrals, claims, and care coordination. To maximize its effectiveness, the directory should include key data elements such as provider and organization names, NPIs, endpoint URLs, supported FHIR versions, capability statements, authentication requirements, practice locations, network affiliations, and verification status. It should also support regular validation and updates and offer APIs for automated integration into clinical and administrative workflows.

Addressing the concern raised by our engineering colleagues, it's clear that the challenge is not just about endpoint mapping but about building an identity-centric architecture for healthcare interoperability. Every entity, provider, organization, device, and patient, needs a persistent, unique digital identity linked to credentials, authorized endpoints, and data sources. This identity-driven model is the only way to ensure that data follows the patient across every care setting and that every agent responsible for care has access to the best possible information. By moving beyond simple endpoint mapping and embracing a holistic, identity-centric approach, we can finally achieve the completeness and reliability needed for optimal value-based care outcomes.

In summary, the path forward for technical standards in VBC lies in harmonizing core frameworks like FHIR, investing in national infrastructure such as validated endpoint directories, and adopting robust digital identity solutions. These steps will reduce complexity and provider burden, foster innovation, and ensure that every participant in the healthcare ecosystem has the tools and information needed to deliver high-quality, patient-centered care.

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